

Lab: Derivation Exercise — Expected Free Energy and Policy Selection

Objective

Derive the Expected Free Energy functional and its decomposition, then apply it to a concrete POMDP to compute policy values.

Part 1: Continuous Action

Goal: Derive the action equation for a continuous system.

Model: Sensory state s depends on action a via $s = h(a) + \omega$ (e.g., $h(a) = a$ for direct motor control). Free energy $F = \frac{1}{2}\pi(s - g(\mu))^2$.

Task:

1. Compute $\partial F / \partial a$ using the chain rule: $\partial F / \partial a = \partial F / \partial s \cdot \partial s / \partial a$
2. Show that $da/dt = -\partial F / \partial a$ drives s toward $g(\mu)$ — action fulfills the sensory prediction
3. Interpret: action is "making predictions come true" through the sensorimotor mapping

Write your response here

Part 2: EFE Decomposition

Goal: Derive the pragmatic-epistemic decomposition of G .

Starting from: $G(\pi) = E_{\{q(o|s, \pi)\}} [\ln q(s|\pi) - \ln p(o|s, \pi)]$

Task:

1. Expand $p(o[s], s[s]) = p(o[s]|s[s]) \cdot p(s[s])$ using the generative model
2. Add and subtract $E_q[\ln q(s[s]|o[s], \pi)]$
3. Show that G decomposes into:
4. **Information gain:** $-E_q[D_{KL}[q(s[s]|o[s], \pi) \parallel q(s[s]|\pi)]]$ (epistemic value)
5. **Pragmatic value:** $E_q[-\ln p(o[s])]$ (how much outcomes deviate from preferences)
6. Explain why information gain is always negative (policies always have non-negative epistemic value)

Write your response here

Part 3: Concrete POMDP Computation

Goal: Compute EFE for a simple POMDP.

Consider a T-maze POMDP with:

- 4 states: {left-reward, right-reward, left-no-reward, right-no-reward}
- 4 observations: {reward, no-reward, left-cue, right-cue}
- 2 actions: {go-left, go-right}
- Preferences: $C = [\ln 2, \ln(1/2), 0, 0]$ (prefer reward, avoid no-reward, neutral on cues)

The A matrix (likelihood):

```
A = [0.9  0.1  0.5  0.5;
      0.1  0.9  0.5  0.5;
      0.5  0.5  0.8  0.2;
      0.5  0.5  0.2  0.8]
```

Task:

1. For policy $\pi_1 = \text{go-left}$ and state belief $q(s) = [0.5, 0.5, 0, 0]$, compute the expected observations $E_q[p(o|s)]$
2. Compute the pragmatic value: $E_q[-\ln p(o)]$ using the C vector
3. Compute the epistemic value: expected information gain
4. Determine which policy (go-left vs go-right) has lower EFE

Write your response here

Part 4: Softmax Policy Selection

Goal: Derive the policy posterior from EFE values.

Given computed EFE values $G(\pi_1), G(\pi_2), \dots, G(\pi_K)$:

Task:

1. Write the softmax: $P(\pi_i) = \exp(-\gamma \cdot G(\pi_i)) / \sum_j \exp(-\gamma \cdot G(\pi_j))$
2. Compute $P(\pi)$ for the T-maze example with $\gamma = 1$ and $\gamma = 16$
3. Show how increasing γ (inverse temperature) makes policy selection more deterministic
4. Discuss: what does $\gamma \rightarrow \infty$ correspond to? What about $\gamma \rightarrow 0$?

Write your response here

Part 5: Synthesis

In 200 words, explain how the EFE framework unifies exploration (epistemic value) and exploitation (pragmatic value) under a single objective, and how this resolves the exploration-exploitation dilemma that plagues classical reinforcement learning.

Write your response here

Lab Summary

Part	Skill Developed	Mathematical Result
1	Continuous calculus	Action equation via chain rule
2	Functional decomposition	$EFE \rightarrow \text{epistemic} + \text{pragmatic components}$
3	Numerical computation	EFE for a concrete POMDP
4	Softmax analysis	Temperature-dependent policy selection
5	Conceptual integration	Exploration-exploitation unification